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EffiCOVID-net: A highly efficient convolutional neural network for COVID-19 diagnosis using chest X-ray imaging

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ABSTRACT

The global COVID-19 pandemic has drastically affected daily life, emphasizing the urgent need for early and accurate detection to provide adequate medical treatment, especially with limited antiviral options. Chest X-ray imaging has proven crucial for distinguishing COVID-19 from other respiratory conditions, providing an essential diagnostic tool. Deep learning (DL)-based models have proven highly effective in image diagnostics in recent years. Many of these models are computationally intensive and prone to overfitting, especially when trained on limited datasets. Additionally, conventional models often fail to capture multi-scale features, reducing diagnostic accuracy. This paper proposed a highly efficient convolutional neural network (CNN) called EffiCOVID-Net, incorporating diverse feature learning units. The proposed model consists of a bunch of EffiCOVID blocks that incorporate several layers of convolution containing (3×3) filters and recurrent connections to extract complex features while preserving spatial integrity. The performance of EffiCOVID-Net is rigorously evaluated using standard performance metrics on two publicly available COVID-19 chest X-ray datasets. Experimental results demonstrate that EffiCOVID-Net outperforms existing models, achieving 98.68% accuracy on the COVID-19 radiography dataset (D_1), 98.55% on the curated chest X-ray dataset (D_2), and 98.87% on the mixed dataset (D_{Mix}) in multi-class classification (COVID-19 vs. Normal vs. Pneumonia). For binary classification (COVID-19 vs. Normal), the model attains 99.06%, 99.78%, and 99.07% accuracy, respectively. Integrating Grad-CAM-based visualizations further enhances interpretability by highlighting critical regions influencing model predictions. EffiCOVID-Net's lightweight architecture ensures low computational overhead, making it suitable for deployment in resource-constrained clinical settings. A comparative analysis with existing methods highlights its superior accuracy, efficiency, and robustness performance. However, while the model enhances diagnostic workflows, it is best utilized as an assistive tool rather than a standalone diagnostic method.

1. Introduction

The emergence of life-threatening viral epidemics like Ebola, Zika, severe acute respiratory syndrome (SARS), and Middle East respiratory syndrome (MERS) is often linked to the evolution of new infectious species or mutations in existing pathogens [1,2]. These viruses have had profound impacts on global health. In December 2019, the world faced another unprecedented challenge with the spread of COVID-19, a novel virus that quickly escalated into a global pandemic. The rapid spread of COVID-19 led to widespread disruption across public health systems, societies, and economies around the globe [3,4].

On March 11, 2020, the World Health Organization (WHO) declared COVID-19 a global pandemic due to its rapid transmission across the world [5]. By August 26, 2024, COVID-19 had resulted in 776.5 million confirmed cases and 7.1 million deaths across 226 countries [6]. The virus's frequent mutations have resulted in the emergence of several variants, raising significant concerns about its persistence and adaptability. While typical symptoms of COVID-19 include cough, fever, loss of taste and smell, and fatigue [7], a substantial number of positive individuals exhibited no symptoms. This asymptomatic transmission complicates efforts to control the virus's spread, making it more challenging to identify and isolate infected individuals. The increase in asymptomatic COVID-19 cases has heavily burdened healthcare systems worldwide, making it more difficult to control the virus [8,9]. Despite the lack of symptoms in many individuals, the virus continues to result in fatali-

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ties, posing ongoing challenges in managing its spread and preventing further outbreaks.

The swift rise in pandemic cases exceeded the capacity of hospitals worldwide, impacting both equipment and personnel. This situation placed immense strain on healthcare systems, underscoring the urgent need for effective methods to rapidly and reliably identify and isolate infected individuals at low cost [10]. The reverse transcription polymerase chain reaction (RT-PCR) test remains the benchmark for identifying COVID-19, as it detects SARS-CoV-2 RNA in respiratory samples [11]. Despite its reliability, the RT-PCR process is often complex and resource-intensive. In light of this, rapid antigen tests (RATs) have emerged as a widely used alternative to expedite diagnosis. However, ensuring the accuracy and reliability of these tests is crucial for effective infection management [12]. Furthermore, the limited availability of diagnostic kits for RT-PCR highlights the necessity for cost-effective solutions that require minimal training for healthcare personnel. Additionally, radiographic assessments, such as chest X-rays and CT scans, are pivotal in identifying respiratory issues linked to COVID-19, even among individuals without noticeable symptoms [13]. Nonetheless, the manual evaluation of these images can be labor-intensive and costly, necessitating specialized training. Therefore, advancing automated detection techniques for COVID-19 is of paramount importance.

Integrating artificial intelligence (AI) in healthcare opens new avenues for the autonomous diagnosis and identification of various infections and diseases [14]. Recent studies indicate that AI-driven techniques, particularly machine learning (ML), have made remarkable progress in bioinformatics and medical image analysis [15]. Within this framework, deep learning (DL) has emerged as a pivotal subset of ML, showcasing significant advancements in medical imaging applications in recent years. It has consistently surpassed many traditional ML techniques, demonstrating superior performance in image recognition and classification tasks [16]. These algorithms can autonomously identify and analyze intricate patterns within large datasets, thus enhancing the efficiency and accuracy of COVID-19 diagnoses. Notable approaches such as convolutional neural networks (CNNs), autoencoders, transfer learning (TL), and ensemble techniques have proven effective in detecting COVID-19 through various imaging modalities. By leveraging the ability of DL to learn complex features from images while minimizing distortions, these methods show promise for accurate detection and classification of the disease. However, despite their potential, several studies have highlighted issues with accuracy in CNN-based detection systems across various medical conditions, including COVID-19, brain cancer, chest cancer, and chickenpox [17]. Additionally, these methods often require extensive training to achieve optimal detection and classification results, which may impact their implementation in clinical practice.

To tackle these challenges, medical professionals need a scalable deep-learning framework to swiftly and accurately identify COVID-19 in X-ray images. This paper introduces EffiCOVID-Net, an innovative CNN with multi-scale feature learning capabilities. This model is specifically designed to effectively extract essential features, even in scenarios where the available chest X-ray data is limited. The proposed architecture has been validated using a publicly available dataset. The EffiCOVID-Net model demonstrates high effectiveness and accuracy in classifying images, thus enhancing the reliability of diagnoses. Furthermore, implementing a Grad-CAM technique allows for adequate visualization of the COVID-19-affected regions within the images, supporting better interpretability of the findings.

The major contribution of this paper include:

- We propose an EffiCOVID-Net model for detecting and classifying COVID-19 using chest X-ray images, employing four multi-scale EffiCOVID blocks for effective learning and preservation.
- Evidence of superior reliability with reduced miss rates even when applied to unfamiliar datasets using EffiCOVID-Net.
- Detailed experimental evaluation on two publicly available COVID-19 chest X-ray datasets, along with a mixed dataset, to confirm the efficacy of EffiCOVID-Net.
- Utilization of Grad-CAM visualizations to assess model predictions visually.
- In-depth comparative analysis between the proposed EffiCOVID-Net model and existing methods, as demonstrated by experimental results.

The rest of the paper is structured as follows: Section 2 discusses related works. Section 3 provides a proposed methodology for COVID-19 detection. Section 4 presents the experiments and results. Section 5 discusses the result of our experiments. Finally, Section 6 concludes the paper.

2. Related works

Early identification of COVID-19 plays a crucial role in mitigating its spread. Chest X-ray and CT images are frequently employed as cost-effective methods for assessing respiratory illnesses globally. Although chest X-rays and CT scans are valuable for identifying COVID-19 pulmonary damage in its infancy, similarities with other viral infections can lead to deception. AI techniques are increasingly applied to the analysis of X-ray and CT images for COVID-19 detection to improve diagnostic precision and reduce reliance on human mistakes. AI has shown substantial potential in diagnosing multiple clinical ailments, which include tumors, malignancy, respiratory infections, and COVID-19. In recent years, advanced AI techniques like DL and ML have shown promising outcomes in medical diagnostics.

Chowdhury et al. [18] conducted two experiments using a chest X-ray dataset for COVID-19 classification. The first experiment categorized the data into two groups: normal and COVID-19, while the second added a third group: viral pneumonia. Transfer learning was applied with and without image augmentation, and their method was tested on eight pre-trained networks. The model achieved 99.41% accuracy in two-class tasks without augmentation, 99.70% with augmentation, and 97.74% accuracy in three-class tasks without augmentation and 97.94% with augmentation. Agarwal et al. [19] proposed a deep CNN for detecting COVID-19 through chest X-ray images. They employed an encoder-decoder architecture for image segmentation, achieving accuracy rates of 94.4% and 95.2% on two distinct datasets. Ozturk et al. [20] presented a deep learning model leveraging the DarkNet architecture, which incorporates seventeen CNN layers and utilizes the Leaky ReLU activation function. The model achieved 98.08% accuracy for binary classification and 87.02% for multi-class classification. Ullah et al. [21] proposed CovidDetNet, an innovative model for COVID-19 detection featuring ten learnable layers, including nine convolutional layers and one fully connected layer. It employs ReLU and Leaky ReLU activation functions and batch and cross-channel normalization. With an accuracy of 98.40% on chest radiograph images, CovidDetNet outperforms existing techniques. Asif et al. [22] developed an automated COVID-19 diagnosis system utilizing a DCNN that processes X-ray images. They employed the Inception-V3 model with transfer learning, analyzing the images without preprocessing. The system achieved a classification accuracy of 96.00%. Ucar et al. [23] introduced an AI-based framework that leverages chest X-ray images to assess COVID-19. They adapted the SqueezeNet architecture for COVID-19 diagnosis using Bayesian optimization for fine-tuning and data augmentation techniques. Their findings indicated that the model achieved a classification accuracy of 98.3% for COVID-19 detection. Aslan et al. [24] introduced a COVID-19 detection model that combines CNN with BiLSTM techniques. This approach yielded a modestly improved 98.14% accuracy by incorporating the more computationally intensive BiLSTM layers alongside the CNN framework.

Several researchers have investigated deep learning (DL) models for early COVID-19 diagnosis, focusing on CNNs applied to chest X-ray

Table 1
Existing Approach for COVID-19 detection using CXR and CT Images.

Author	Classes	Techniques	Datasets	Metrics
Chowdhury et al. [18]	2 and 3	Image augmentation (IA) with TL	Public	With IA: 99.70% for binary, and 97.94% for multi classes
Ozturk et al. [20]	2 and 3	Dark-Net	Public	Accuracy: 98.08% for binary and 87.02% for multi classes
Ullah et al. [21]	3	CovidDetNet	Public	Accuracy: 98.40%, F1Score: 96.82%, Precision: 98.52% and Recall: 97.55
Ashif et al. [22]	3	DCNN	Public	Accuracy: 96.00
Ucar et al. [23]	3	Deep Bayes-Squeeznet	Public	Accuracy: 98.26, and F1 Score: 98.25%
Aslan et al. [24]	3	CNN+BiLSTM	Public	Accuracy: 98.14%, F1: 98.20%, Recall: 98.26%, and Precision: 98.16%
Ibrahim et al. [27]	3	Multi-Classification	Public	AUC: 99.6%, Accuracy: 98.05%, Sensitivity: 98.43%, and Specificity: 99.5%
Nayak et al. [29]	3	LW-CORONet	Public	Accuracy: 95.67%, Precision: 95.67%, Sensitivity: 97.00%
George et al. [30]	3	GrayVIC model	Public	Accuracy: 97.41%, Sensitivity: 94.52%
Sethy et al. [31]	3	ResNet50 plus SVM	Public	Accuracy: 95.33%, Sensitivity: 95.33%

images. Studies such as those by [20,21,25–30] have addressed key challenges, including the limitations of small datasets, distinguishing COVID-19 cases from chest X-ray images, and employing ensemble techniques combined with cross-dataset analysis. A comparative summary of the performance of these state-of-the-art models is provided in Table 1.

Previous research has encountered several challenges, including unreliable detection, small datasets, biases, and using CNNs without adequate image preprocessing. Consequently, there is a strong need for a CNN capable of effectively learning significant features from limited training data. Additionally, validating convolutional neural network models with small datasets can significantly affect their performance. To address these challenges, this paper employs various image preprocessing techniques. It introduces the EffiCOVID-Net model, which adeptly captures diverse characteristics from chest X-ray images of a small size.

3. Methodology

This section outlines the proposed EffiCOVID-Net architecture designed to enhance COVID-19 detection. Fig. 1 provides an abstract overview of the EffiConv-Net model, which operates through several stages. Initially, the input images are categorized into three distinct classes: COVID-19, viral pneumonia, and normal images. The images then undergo preprocessing techniques before advancing to the subsequent stage. Data augmentation techniques are applied to expand the dataset and improve the model's generalization capabilities, artificially increasing the image data's diversity and volume. Next, features are extracted from the preprocessed images using advanced feature extraction methods. The model then undergoes rigorous training and validation, ensuring it performs well on unseen data. Finally, the trained model is used for image classification, where it analyzes the extracted features to predict the presence of COVID-19, viral pneumonia, or normal conditions in the input images.

3.1. Dataset

The proposed EffiCOVID-Net model for COVID-19 detection exercises X-ray images, which undergo tailored adjustment using specific parameters.

3.1.1. X-ray dataset

The proposed model utilizes X-ray images from two primary sources. The first dataset, denoted as D_1, comprises 3,616 COVID-19 images, 10,192 normal X-ray images, and 1,345 viral pneumonia images sourced from the COVID-19 Radiography Database [18,32]. The second dataset, referred to as D_2, includes 1,281 COVID-19 X-rays, 3,270 normal X-ray images, and 1,656 viral pneumonia X-ray images from the Curated Chest X-ray Image Dataset [33]. By amalgamating these datasets, we construct a comprehensive dataset named D_{Mix}, which encompasses 4,897 COVID-19 images, 13,462 normal images, and 3,001 viral pneumonia images. This extensive dataset serves as a robust foundation for the training and evaluation of the proposed model. Fig. 2 illustrates a sample of the X-ray images utilized in this paper. Additionally, Fig. 3 depicts the datasets employed. Fig. 4 presents the distribution of image classes within (a) D_1, (b) D_2, and (c) D_{Mix} datasets.

3.2. Image preprocessing

In the subsequent stage, images undergo preprocessing, which is essential for transforming raw data into formats suitable for machine learning (ML) and deep learning (DL) approaches. This step enhances image quality and increases the chances of successful assessments. Since the images come from various sources, there are differences in size and quality across the dataset [34]. Despite efforts to reduce overfitting, it remains a potential issue in DL models. This risk can be mitigated by limiting the number of trainable parameters, using early stopping techniques, and applying data augmentation to smaller datasets. Standardizing the dataset involves resizing all input images to 256×256 pixels, employing data transformation and feature engineering techniques, and using normalization methods to stabilize the gradient descent process. Fig. 5 shows X-ray images after preprocessing.

3.3. Data normalization

The next stage in the proposed detection framework underscores the importance of data normalization for developing precise models in deep learning (DL) algorithms. This stage employs several normalization techniques, including Min-Max scaling, decimal normalization, and mean and standard deviation normalization, to transform pixel values to a range between zero and one [35]. To categorize the target variable, COVID-19, into distinct classes, the "OneHot Encoder" function from the Python standard library is used [36]. Specifically, as detailed in Table 2, COVID-19 is encoded as 0, normal as 1, and viral pneumonia as 2.

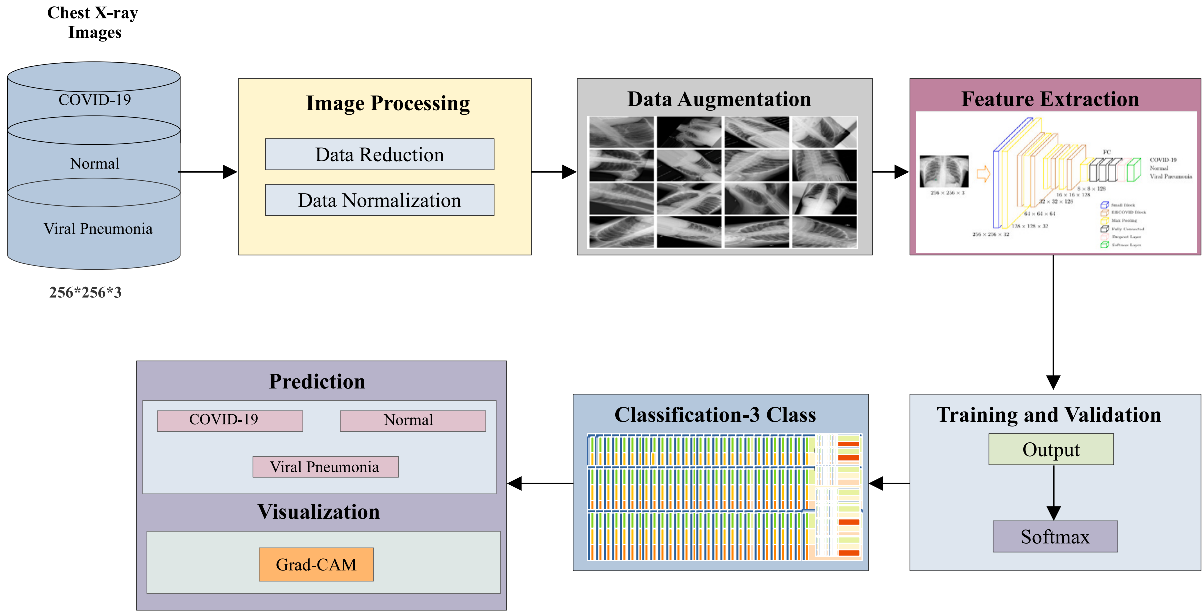
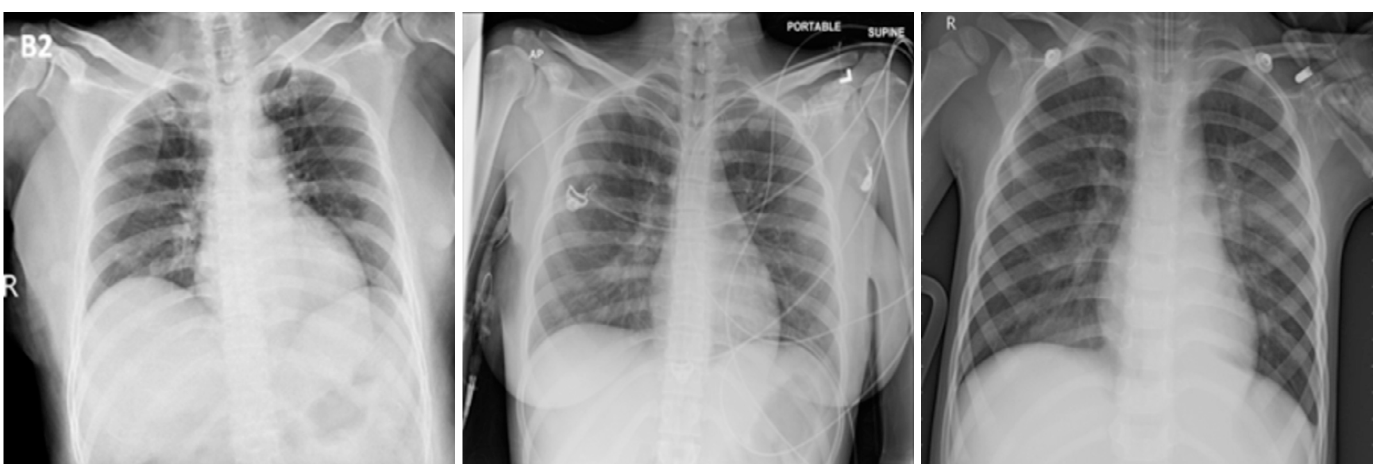


Fig. 1. Abstract Overview of Proposed Model.



(a) COVID-19 (b) Normal (c) Viral Pneumonia

Fig. 2. Sample Images.

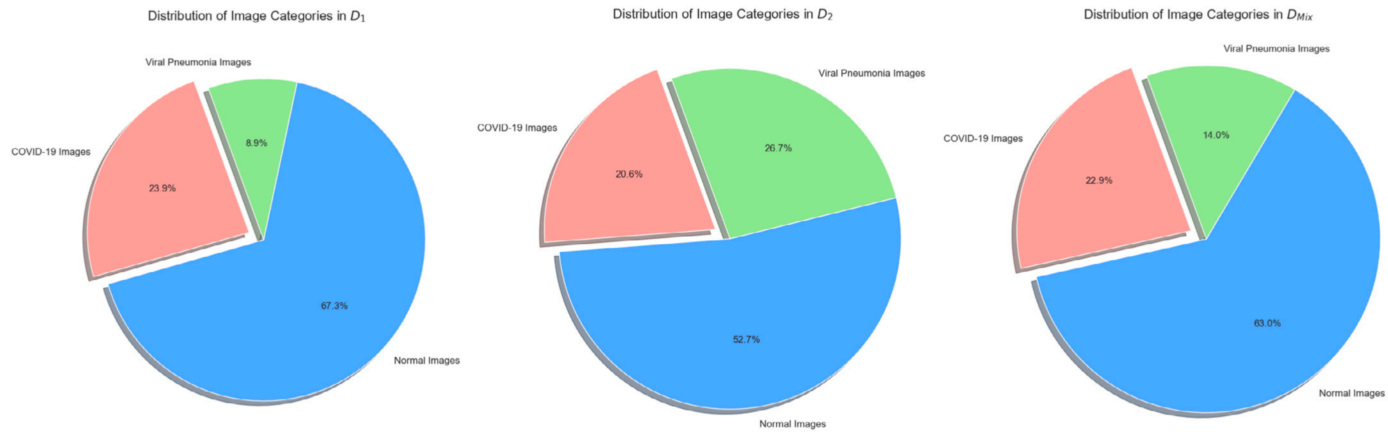
Fig. 3. Description of Dataset Used in this Study.

Table 2
Assigning Class Labels to the D_1, D_2 and D_{Mix} datasets.

	D_1, D_2 and D_{Mix} Dataset		
Classes	COVID-19	Normal	Viral Pneumonia
Set	0	1	2

3.4. Data augmentation

Data augmentation enhances the performance of ML models by increasing the training dataset with modified versions of existing images, thereby expanding the overall image count [37]. This approach simplifies the process of obtaining additional images for dataset enhancement. Techniques such as oversampling and data wrapping introduce greater image diversity. Various augmentation methods mitigate overfitting, including channel shifting, mirroring, rotation, inversion, zooming, and shearing. These methods preserve the essential features of the images while adjusting angles to correct perspectives and introducing controlled distortion by varying pixel density.



(a) (b) (c)

Fig. 4. Distribution of Images Classes in (a) D_1 (b) D_2 (c) D_{Mix} Dataset.

(a) COVID-19 (b) Normal (c) Viral Pneumonia

Fig. 5. Post Preprocessing Stage.

3.5. Feature extraction

Feature extraction is crucial for improving the effectiveness, accuracy, and interpretability of DL and ML systems [38]. The proposed EffiCOVID-Net model consists of four EffiCOVID blocks, as illustrated in Fig. 6. The structure of the proposed model and the details of its individual blocks are described below.

3.5.1. Small block

The EffiCOVID block comprises multiple smaller units, each consisting of three layers, referred to as α_{small}. Each α_{small} block includes a Conv2D, Batch Normalization (BN), and ReLU activation layers. The Conv2D layer utilizes a 3×3 filter to extract detailed features from the X-ray images while maintaining a padding value of "same" to ensure consistent feature map sizes across the EffiConv block. In the proposed model, let $\alpha^{K}_{small}(M)$ denote the output of a mini block with K filters. Here, $\theta^{fs,fs,K}(M)$ represents the convolution operation with K filters of size $fs \times fs$, β denotes the Batch Normalization (BN) function, and γ denotes the ReLU activation function. The Equation (1) formulation of the small unit is given by:

$$\alpha^{small}_{K}(M) = \beta\left(\theta^{3,3,K}(M)\right) \tag{1}$$

3.5.2. EffiCOVID block

The EffiCOVID block is a multi-scale feature extraction component integrated into the EffiCOVID-Net architecture (Fig. 7). It operates in four stages to capture features at varying receptive field sizes using different filters. Smaller filters, such as 3×3, are utilized to detect fine details, while larger filters are used to extract more general features like the shape of the lung regions. To enable the extraction of multi-scale features, filters of different sizes are applied in parallel to the input data, and the resulting feature maps are concatenated. Since larger filters require more parameters, multiple Conv2D layers with 3×3 kernels are employed to achieve similar feature extraction capabilities. The following mathematical formulation computes the size of the receptive field.

Let X_{in} represent the dimensions of the input feature map, X_{out} denote the dimensions of the output feature map, and k indicate the kernel size. p the padding size, and s the stride. The size of the output feature map resulting from a convolution operation can be mathematically described by:

$$X_{out} = \left\lfloor \frac{X_{in} + 2p - k}{s} \right\rfloor + 1 \tag{2}$$

Let m denote the "shift", which represents the cumulative effect of strides accumulated from all previous layers. The shift in the n n-th layer can be mathematically defined as:

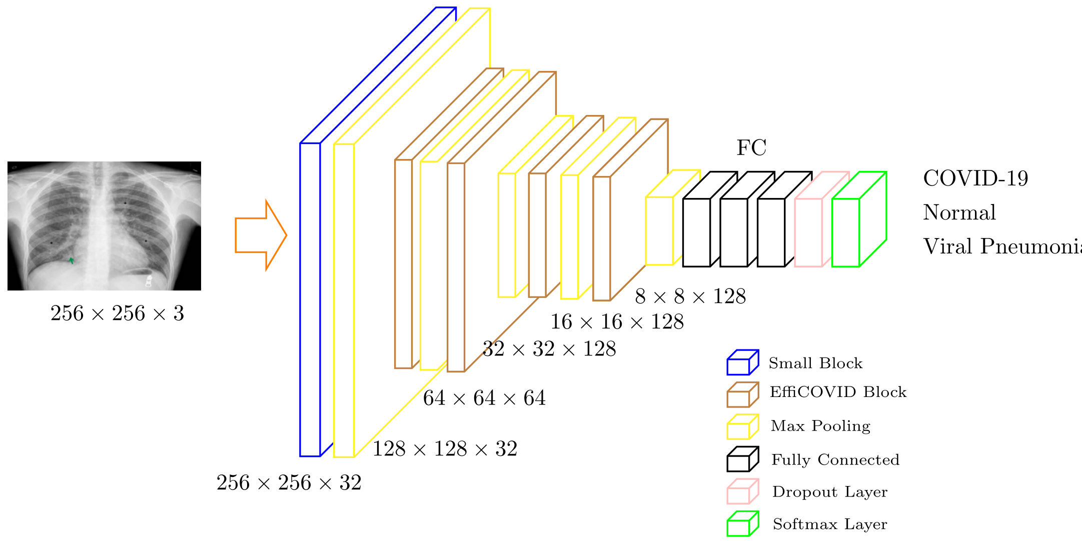


Fig. 6. Proposed EffiCOVID-Net Model.

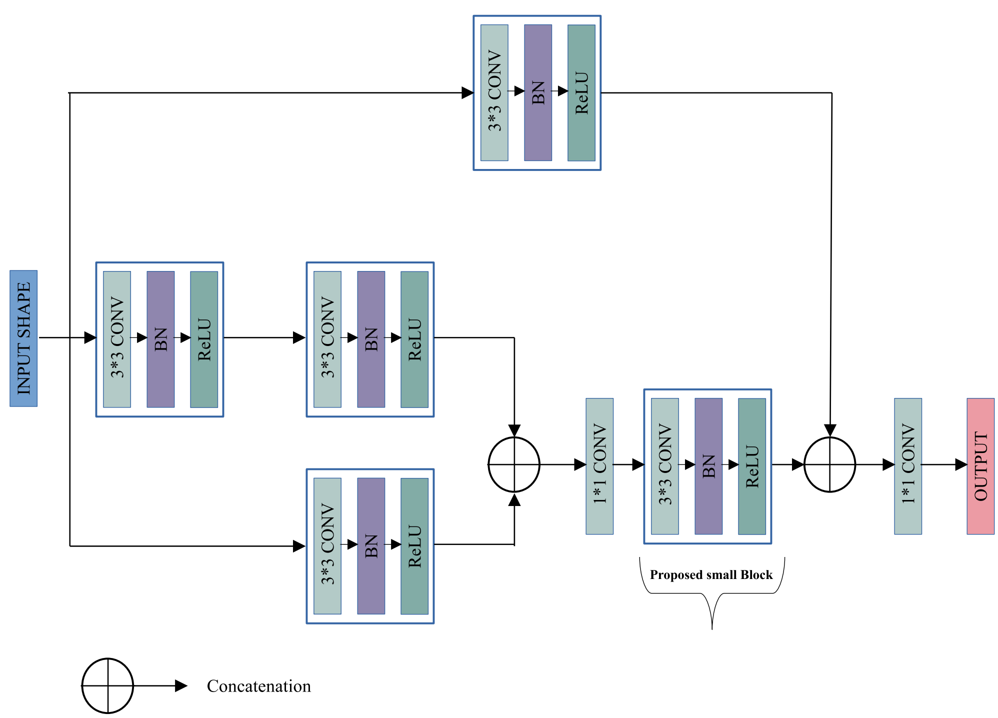


Fig. 7. Proposed EffiCOVID Block.

$$m_n = m_{n-1} \times s \tag{3}$$

The size of the receptive field can be determined using the following formula:

$$r_n = r_{n-1} + (l-1) \times m_{n-1} \tag{4}$$

where r_n represents the receptive field size of the n n-th layer. This equation can be expressed in a generalized form as:

$$r_n = r_{n-1} + (l-1) \times \sum_{j=1}^{n-1} s_j \tag{5}$$

The EffiCOVID block performs feature extraction with receptive fields of sizes 3 and 5 through two parallel processing streams. A single small unit is used in the first stream, while the second stream utilizes two stacked small units. The extracted features are merged via concatenation, maintaining receptive fields of sizes 3 and 5. This combined output is then processed by a 1×1 Conv2D layer, which reduces the feature map channels from $k \times k \times fs$ to $k \times k \times (fs/2)$. An additional small unit processes the resulting feature maps to expand the receptive fields to sizes 5 and 7. The final features are concatenated with the original input to the EffiCOVID block, facilitating the extraction of multi-scale features with receptive field sizes 3, 5, and 7. The concatenated output undergoes channel reduction via a 1×1 Conv2D layer. The operation of the EffiCOVID block can be mathematically expressed as:

$$\alpha' = \alpha^K_{small}\left(\alpha^K_{small}(M)\right) \oplus \alpha^K_{small}(M) \tag{6}$$

$$\alpha^K_{EffiCOVID}(M) = \theta^{1,1,K}\left(\alpha^K_{small}\left(\theta^{1,1,K}(\alpha')\right) \oplus \alpha^K_{small}(M)\right) \tag{7}$$

where $\alpha^K_{EffiCOVID}(M)$ represent the output of the EffiConv block ap-

Table 3
EffiCOVID-Net Model Summary.

Layer Name	Details	Output Shape
Small Block	[3×3 Conv2D] × 1	256×256 × 32
Max Pooling	[2 ×2 max pool, s = 2]	128×128 × 32
EffiCOVID Block (1)	[1×1 Conv2D] × 2 × 5	128×128 × 32
Max Pooling	[2 ×2 max pool, s = 2]	64×64 × 32
EffiCOVID Block (2)	[1×1 Conv2D] × 2 × 5	64×64 × 64
Max Pooling	[2 ×2 max pool, s = 2]	32×32 × 64
EffiCOVID Block (3)	[1×1 Conv2D] × 2 × 5	32×32 × 128
Max Pooling	[2 ×2 max pool, s = 2]	16×16 × 128
EffiCOVID (4)	[1×1 Conv2D] × 2 × 5	16×16 × 128
Max Pooling	[2 ×2 max pool, s = 2]	8×8 × 128
Flatten	8192	1×1 × 128
Dense	256	1×1 × 256
FCL	3-D Fully Connected (Dropout: 0.4), Softmax	1×1 × 3

plied to input M with K filter and $\oplus$ denote the concatenation operation (Table 3).

3.5.3. Overall proposed EffiCOVID-net model

EffiCOVID-Net is a streamlined model designed to learn features across multiple scales efficiently. The architecture starts with a compact block that initiates feature learning, followed by a max-pooling layer that reduces the feature map's size. The model then utilizes four EffiCOVID blocks to perform multi-scale feature extraction, with each block followed by a max-pooling layer. Afterward, the features are processed by a pooling, flattening, dense, and dropout layer. The network concludes with a classification layer comprising three neurons, using softmax activation to distinguish between COVID-19, Normal, and Viral Pneumonia in X-ray images.

Algorithm 1 EffiCOVID-Net for Automated COVID-19 Detection.

Require: Chest X-ray dataset D with labeled images, Image resolution (H, W), Number of classes C, Hyperparameters: learning rate α, batch size B, epochs E

Ensure: Trained EffiCOVID-Net model with optimized classification performance, Evaluation results including accuracy, precision, recall, and F1-score, AUC Score

Step 1: Data Preprocessing
- Load dataset D containing COVID-19, normal, and pneumonia images
- Resize each image to a fixed resolution (H, W)
- Normalize pixel values to range [0,1]
- Apply data augmentation (rotation, flipping, contrast enhancement)
- Split dataset into training D_{train}, validation D_{val}, and testing D_{test} subsets

Step 2: Model Initialization
- Initialize EffiCOVID-Net with a lightweight convolutional architecture
- Define EffiCOVID blocks with:
  - Convolutional layers using 3×3 filters
  - Recurrent connections for enhanced feature extraction
  - Batch normalization and ReLU activation
  - Max pooling for dimensionality reduction
- Integrate multi-scale feature learning via parallel convolutional branches
- Add fully connected layers with dropout and softmax activation for classification

Step 3: Model Training
for epoch E in range 1 to N **do**
  for each mini-batch B from D_{train} **do**
    Perform forward pass through EffiCOVID-Net
    Compute loss L (categorical cross-entropy)
    Update model parameters using backpropagation and Adam optimizer
  end for
  Validate model on D_{val} and adjust hyperparameters if necessary
end for

Step 4: Model Evaluation
- Test trained model on D_{test}
- Compute performance metrics: Accuracy, Precision, Recall, F1-score, AUC Score
- Generate Grad-CAM heatmaps for model explainability

Table 4
Settings of Variables used in this study.

Variables	D_1, D_2, and D_{Mix} (X-ray) Dataset	
	Initial Variable	Optimized Variable
Batch Size	8, 16, 32, 64	32
Learning Rate	.0001, .001, .01. 0.1	.001
Epochs	30, 40, 50, 70	60

4. Experiments and results

This section assesses the effectiveness of the proposed EffiCOVID-Net architecture for COVID-19 detection. It details the implementation setup, evaluation metrics, and experimental findings of the EffiCOVID-Net model on the D_1, D_2, and D_{Mix} datasets. The datasets are split into an 80:20 ratio of X-ray images for training and testing, corresponding to 80% training, 10% testing, and 10% validation for 3 class and 2 class classifications. The section also presents a comparative analysis against pre-trained ImageNet models and existing methodologies, highlighting the performance of EffiCOVID-Net.

4.1. Implementation setup

The proposed EffiCOVID-Net has been evaluated using the D_1, D_2, and D_{Mix} X-ray datasets, designed to capture a range of COVID-19-related diagnostic scenarios. To combat overfitting and ensure robust generalization across unseen data, data augmentation techniques are applied to the training sets of each dataset. The model's architecture concludes with a classification layer comprising three neurons; each is activated using the softmax function to handle the multi-class output. Categorical cross-entropy has been selected as the loss function due to its suitability for classification tasks with mutually exclusive classes [39]. The study underscores the significance of fine-tuning hyperparameters to achieve optimal machine-learning model performance [40]. Iteratively refining key parameters such as epochs, batch size, and learning rate, a grid search strategy is employed to streamline the process, reducing the need for time-consuming manual adjustments. The initial and optimal parameters used during the experiments are detailed in Table 4. The proposed model is trained and tested on an 11th Gen Intel Core i7-11700 processor, 32GB of RAM, and an NVIDIA Tesla V100-SXM2-12GB GPU, operating at 2.50GHz. The model uses the "Adam optimizer" to maintain consistent performance across categories. The experiments use the Keras framework with TensorFlow, and intensity normalization is applied to ensure numerical stability [41]. The datasets are split into 80% for training, 10% for validation, and 10% for testing, as detailed in Table 5.

4.2. Evaluation metrics

The performance of the proposed model has been evaluated using several key metrics, including accuracy (Acc), precision (Pre), recall (Rec), F1-score (F1), and area under-the-curve-receiver operating characteristic (AUC-ROC), which are widely used in biomedical imaging research [42,43]. These metrics have been extensively employed in COVID-19, normal, and pneumonia detection studies, demonstrating their reliability in evaluating deep learning-based diagnostic models [44,45]. The computation of these metrics relies on fundamental classification outcomes: false positives (FP), true positives (TP), false negatives (FN), and true negatives (TN).

Table 5
X-ray Dataset used by EffiCOVID-Net Model.

Set	D_1 Dataset			D_2 Dataset			D_{Mix} Dataset		
	COVID-19	Normal	Viral Pneumonia	COVID-19	Normal	Viral Pneumonia	COVID-19	Normal	Viral Pneumonia
Train	2892	8153	1076	1,024	2,616	1,324	3,917	10769	2,400
Valid	361	1019	134	128	327	165	489	1346	300
Test	363	1020	135	129	327	167	491	1347	301
Total	**3,616**	**10,192**	**1,345**	**1,281**	**3,270**	**1,656**	**4,897**	**13,462**	**3,001**

Accuracy measures the overall correctness of the model by calculating the ratio of correctly predicted instances to the total number of instances.

$$\text{Acc} = \frac{TP + TN}{TP + TN + FP + FN} \tag{8}$$

Precision quantifies the proportion of correctly predicted positive instances out of all instances predicted as positive.

$$\text{Pre} = \frac{TP}{TP + FP} \tag{9}$$

Recall measures the ability of the model to identify positive instances out of all actual positive instances correctly.

$$\text{Rec} = \frac{TP}{TP + FN} \tag{10}$$

The F1-score is the harmonic mean of precision and recall, balancing the two.

$$\text{F1} = 2 \times \frac{\text{Pre} \times \text{Rec}}{\text{Pre} + \text{Rec}} \tag{11}$$

AUC-ROC evaluates the model's ability to distinguish between classes, with the area under the curve as a plot of the true positive rate versus the false positive rate at various threshold settings.

$$\text{FPR} = \frac{FP}{FP + TN} \tag{12}$$

4.3. Results

The proposed model's effectiveness and learning capability are thoroughly evaluated by conducting extensive training and testing on diverse patient samples from the D_1, D_2, and D_{Mix} X-ray datasets. The 80:20 split ratio is used for training, validation, and testing. In the initial phase, experiments on the D_1 dataset utilized 15,153 X-ray images, with 12,121 allocated for training, 1,514 for validation, and 1,518 for performance assessment. In the second phase, experiments on the D_2 dataset involved 6,207 X-ray images, divided into 4,964 for training, 620 for validation, and 623 for performance assessment. For the combined D_{Mix} dataset, 21,360 X-ray images were used, with 17,086 for training, 2,135 for validation, and 2,139 for testing. Tables 6 and 7 provide a comprehensive assessment of the proposed model, showcasing its precision, recall, F1 score, accuracy, and AUC-ROC, along with the associated confusion matrix for 3 class and 2 class classifications shown in Fig. 9 and 12.

4.3.1. Three-class classifier

The experiment is repeated to optimize hyperparameters, including batch size, node size, drop rate, and learning rate for X-ray images. Fig. 8 shows the training curves of the EffiCOVID-Net model on the D_1, D_2, and D_{Mix} datasets, demonstrating strong convergence within 60 epoch. The proposed model achieved an accuracy of 98.68%, 98.55%, and 98.87% on D_1, D_2, and D_{Mix} X-ray datasets, with detailed classification results provided in Table 6. Fig. 9 illustrates the confusion matrices produced by the EffiCOVID-Net model on the test set of the X-ray datasets. The D_1 dataset reveals that 9 out of 363 COVID-19 samples are incorrectly classified as normal, 10 out of 1,020 normal samples are misclassified as either COVID-19 or viral pneumonia, and 6 out of 135 viral pneumonia samples are incorrectly classified as normal. While on the D_2 X-ray dataset, 1 out of 129 COVID-19 samples is wrongly classified as normal, 1 out of 327 normal samples is incorrectly labeled as COVID-19, and 7 out of 167 viral pneumonia samples are misclassified as either COVID-19 or normal. In the D_{Mix} datasets, 7 out of 491 COVID-19 samples are wrongly classified as normal, 12 out of 1347 normal samples are incorrectly labeled as either COVID-19 or viral pneumonia, and 5 out of 301 viral pneumonia samples are misclassified as either COVID-19 or normal. The proposed model achieves AUC-ROC values of 99.94%, 99.89%, and 99.95% for the D_1, D_2, and D_{Mix} imaging datasets. These high AUC-ROC values indicate the model's exceptional ability to distinguish between COVID-19, viral pneumonia, and normal cases. This performance is further illustrated in Fig. 10, which showcases the ROC curves for each dataset, highlighting the model's robustness and reliability in accurately classifying the images across all datasets.

Table 6
Results of Three-Class Classification with the Proposed EffiCOVID-Net Model.

Parameters	D_1	D_2	D_{Mix}
Accuracy (%)	98.68	98.55	98.87
AUC (%)	99.94	99.89	99.90
Precision			
COVID-19	97.54	98.46	97.98
Normal	99.21	97.89	99.18
Viral Pneumonia	97.77	100	98.99
Recall			
COVID-19	98.62	99.22	98.58
Normal	98.82	99.69	99.10
Viral Pneumonia	97.77	95.80	98.33
F1-score			
COVID-19	98.07	98.83	98.27
Normal	99.01	98.78	99.13
Viral Pneumonia	97.77	97.85	98.65

Table 7
Results of Two-Class Classification with the Proposed EffiCOVID-Net Model.

Parameters	D_1	D_2	D_{Mix}
Accuracy (%)	99.06	99.78	99.07
AUC (%)	99.94	100	99.94
Precision			
COVID-19	97.55	100	97.97
Normal	99.60	99.69	99.47
Recall			
COVID-19	98.89	99.22	98.57
Normal	99.11	100	99.25
F1-score			
COVID-19	98.21	99.60	98.26
Normal	99.35	99.84	99.35

4.3.2. Two-class classifier

Fig. 11 shows the accuracy and loss curves for the proposed model on the D_1, D_2, and D_{Mix} dataset over 60 training epochs for the two-

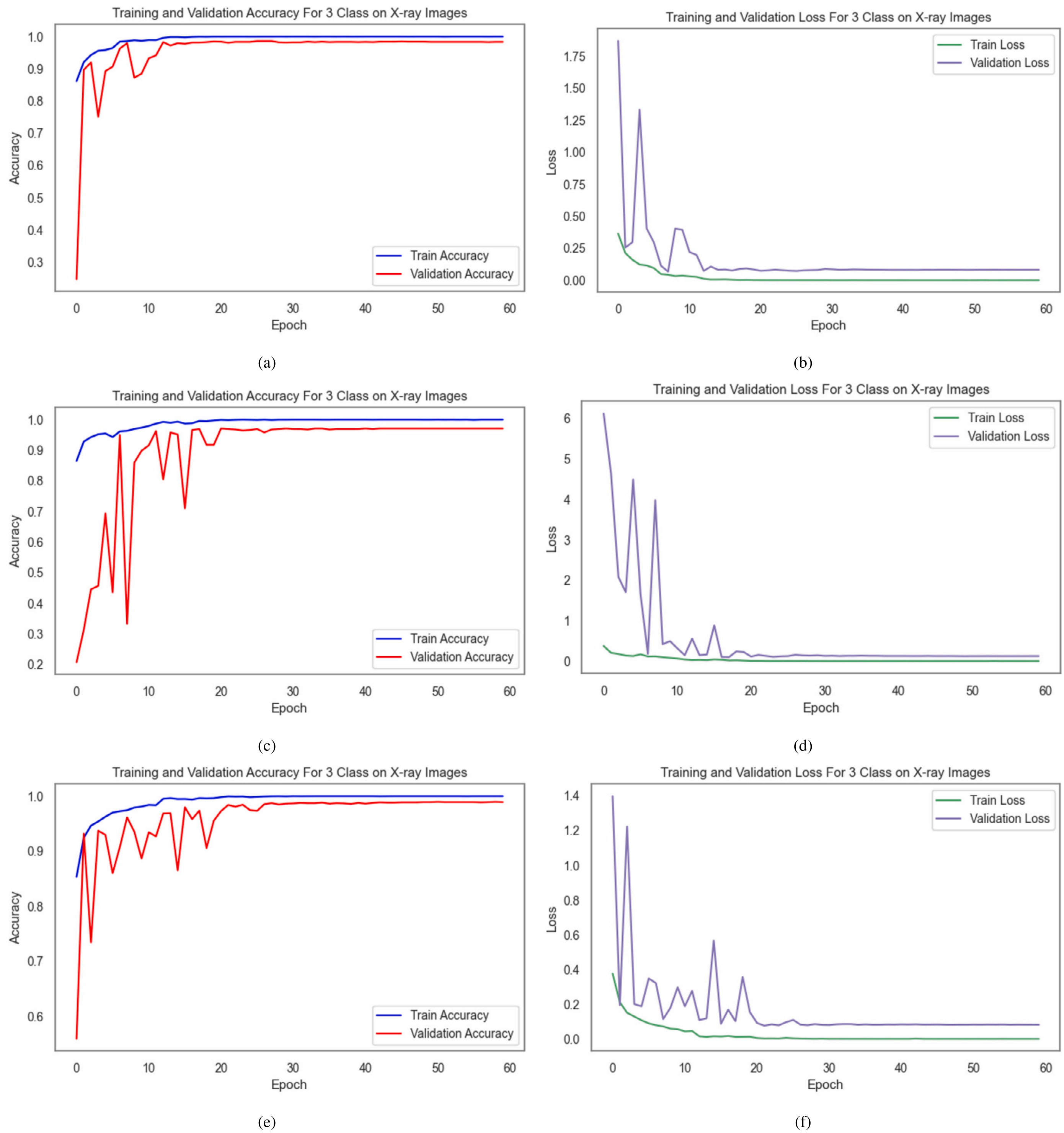
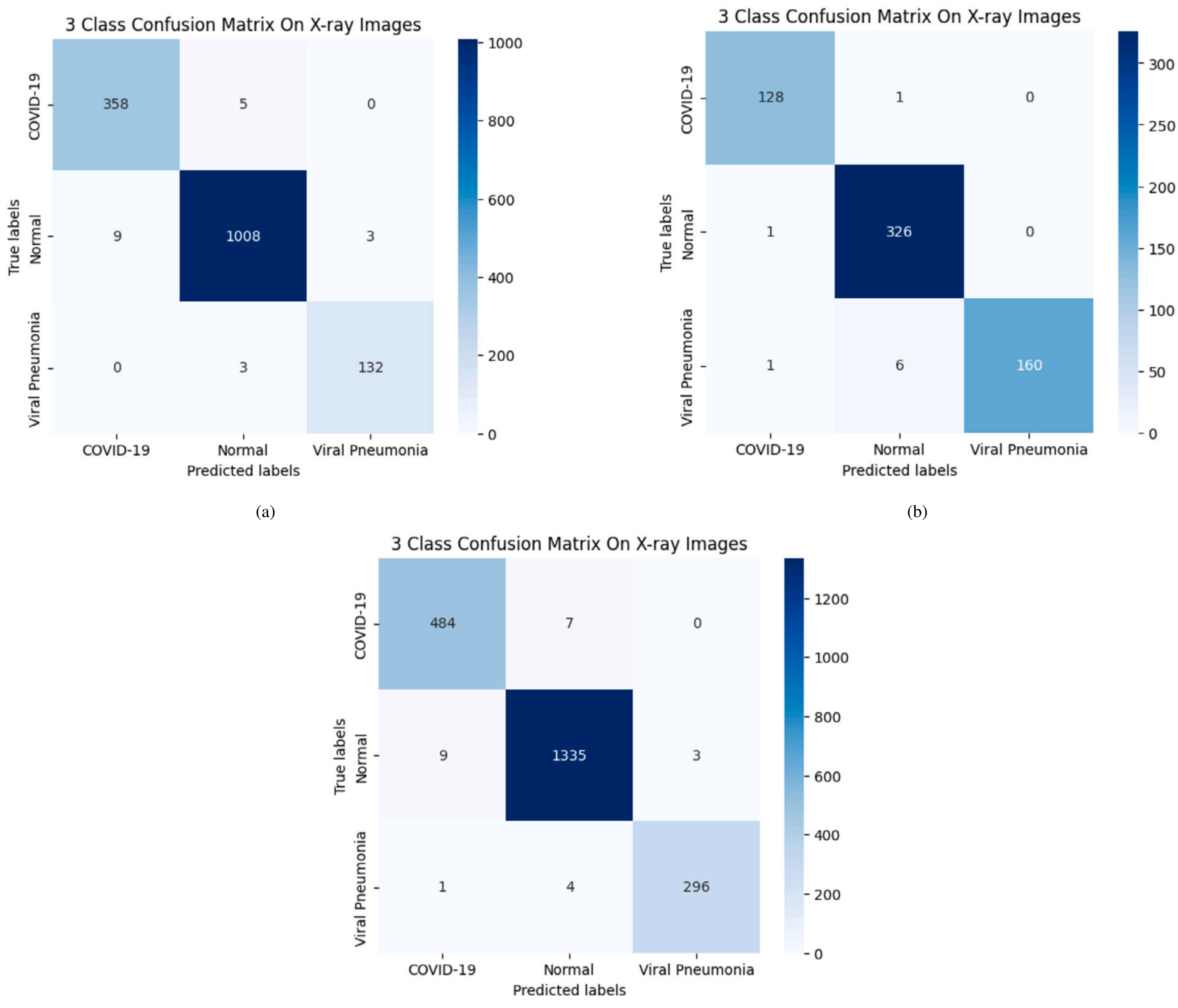


Fig. 8. Training Curves for 3 class classification (accuracy and loss vs. epoch) on (a) and (b) for D_1, (c) and (d) for D_2, and (e) and (f) for D_{Mix} dataset.

class classification. The model achieved high accuracy rates of 99.06%, 99.78%, and 99.07% on various X-ray datasets, as detailed in Table 7. Fig. 12 illustrates the confusion matrices for the two class classifications (COVID-19 and Normal) on the D_1, D_2, and D_{Mix} datasets, using the proposed model, which demonstrated superior performance compared to other models. In the D_1 dataset, out of 363 COVID-19 images, 359 were correctly classified, resulting in 4 misclassifications as normal. For the normal class, 1,011 out of 1,015 images were accurately classified, yielding 4 false positives. Similarly, the D_2 dataset reveals that 128 out of 129 COVID-19 images were correctly identified, with only 1 misclassified as normal. In contrast, the normal class achieved perfect accuracy, classifying all 327 images correctly with zero false positives. In the D_{Mix} dataset, 484 out of 491 COVID-19 images were correctly classified, leading to 7 misclassifications as normal. For the normal class in this dataset, 1,337 out of 1,347 images were accurately classified, resulting in 10 false positives. Overall, these results highlight the effectiveness of the proposed model across all datasets, demonstrating its robust performance in accurately distinguishing between COVID-19 and



(a) (b) (c)

Fig. 9. Confusion Matrices for 3 Class Classification on (a) D_1, (b) D_2, and (c) D_{Mix} dataset.

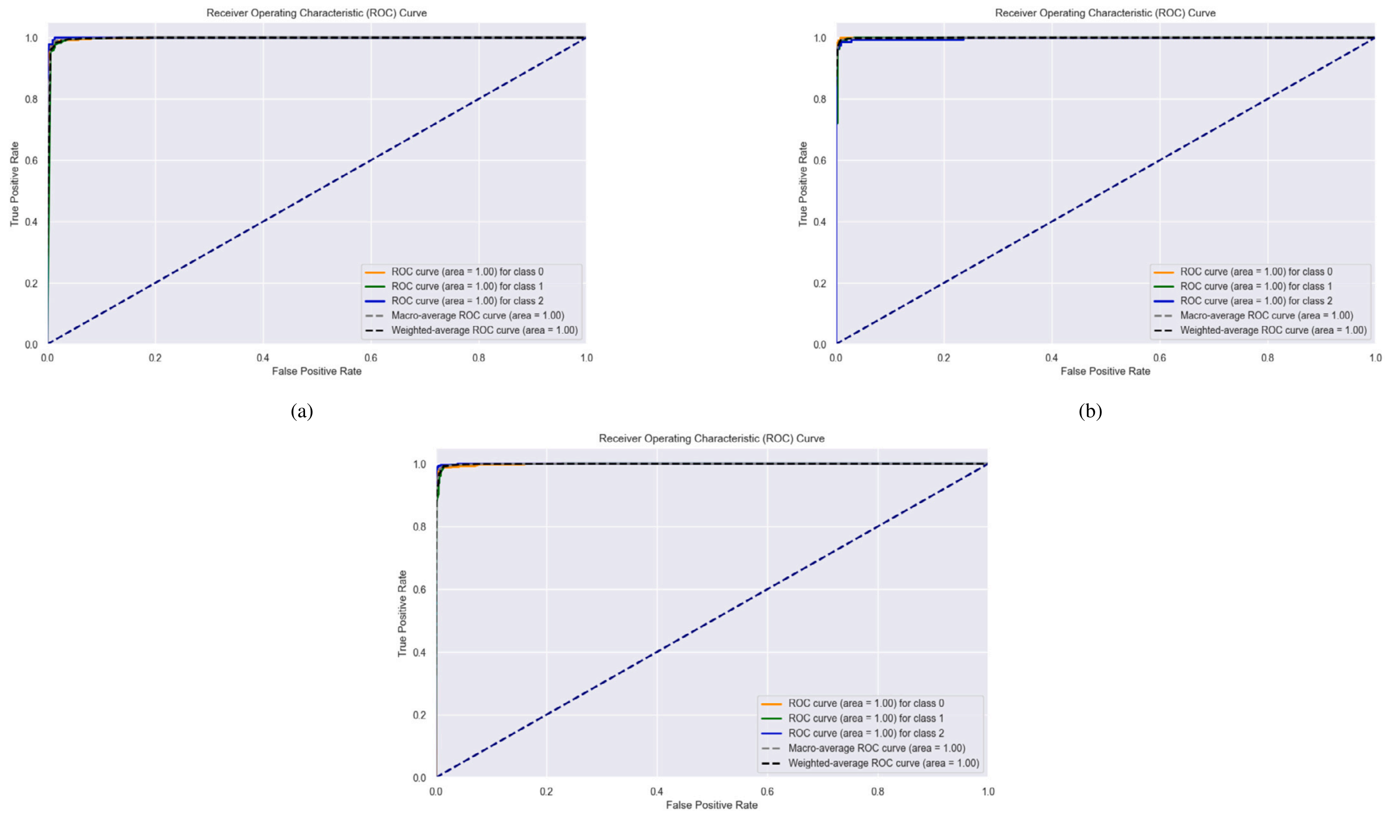
normal cases. The proposed model achieves impressive AUC-ROC values of 99.94%, 100%, and 99.94% for the D_1, D_2, and D_{Mix} datasets, respectively. These high AUC-ROC values indicate the model's exceptional ability to distinguish between COVID-19 and normal cases, as shown in Fig. 13.

4.4. Comparison with pre-trained model

The study compared the performance of ImageNet pretrained CNN architectures, including Inception V3 [46], DenseNet-161 [47], ResNet-101 [48], MobileNet V2 [49], VGG-16 [50], EfficientNet B0 [51], and Xception [52] on D_1, D_2, and D_{Mix} X-ray datasets and compared their results with the proposed model. Each model's results were evaluated against the proposed model. Notably, many of these architectures are parameter-heavy, with Inception V3 comprising 24 million parameters and VGG-16 reaching 136 million. In contrast, the proposed model contains only 10 million parameters, offering significantly improved memory efficiency and highly suitable for real-time COVID-19 detection using X-ray images. Tables 8, 9, and 10 present the classification results, which clearly show that the proposed model outperforms all the compared ImageNet models on the D_1, D_2, and D_{Mix} X-ray datasets, achieving comparable or superior performance in all metrics. These findings underscore the potential of the proposed model for practical deployment in resource-constrained environments.

Table 8
Performance Comparison of Pretrained CNN Models on the D_1 Dataset.

Model	Pre	Rec	F1	AUC	Acc
Inception-V3	97.55	97.91	98.23	98.69	97.42
DenseNet161	97.92	98.11	98.01	98.92	97.89
ResNet101	98.01	98.23	98.11	99.07	98.03
MobileNet-V2	98.19	97.46	97.82	99.17	98.31
VGG-16	98.09	98.35	98.21	98.83	98.29
EfficientNet-B0	97.13	97.27	97.19	98.09	97.93
Xception	98.06	98.26	98.15	99.32	98.47
Proposed	98.17	98.40	98.29	99.94	98.68



(a) (b) (c)

Fig. 10. AUC-ROC Curves for 3 class classification on (a) D_1, (b) D_2, and (c) D_{Mix} dataset.

Table 9
Performance Comparison of Pretrained CNN Models on the D_2 Dataset.

Model	Pre	Rec	F1	AUC	Acc
Inception-V3	97.05	97.12	97.08	96.38	97.19
DenseNet161	96.29	96.11	96.19	95.92	96.83
ResNet101	98.32	98.27	98.29	98.69	98.13
MobileNet-V2	97.92	98.10	98.00	98.92	97.79
VGG-16	97.21	97.10	97.65	97.74	96.37
EfficientNet-B0	96.81	97.39	97.09	98.19	96.85
Xception	98.27	98.13	98.19	98.70	97.62
Proposed	98.78	98.23	98.49	99.89	98.55

Table 10
Performance Comparison of Pretrained CNN Models on the D_{Mix} Dataset.

Model	Pre	Rec	F1	AUC	Acc
Inception-V3	98.11	97.43	97.76	98.63	97.99
DenseNet161	97.75	98.23	97.98	98.31	97.36
ResNet101	97.39	97.62	97.50	97.76	97.01
MobileNet-V2	98.12	97.91	98.01	98.39	98.07
VGG-16	97.91	97.43	97.66	98.19	97.82
EfficientNet-B0	98.01	98.15	98.07	98.36	97.65
Xception	98.29	98.45	98.36	98.74	98.06
Proposed	98.71	98.67	98.68	99.90	98.87

4.5. Grad-CAM

The proposed model outperformed ImageNet-pretrained CNN models and state-of-the-art deep learning techniques. To gain further insights into the model's behavior and understand the features it emphasizes, we applied Grad-CAM visualizations [53]. These visual interpretations help validate the model's predictions and provide a clearer understanding of whether it is learning meaningful features from the images or focusing on irrelevant background details. This step is crucial for ensuring the model's decision-making process is based on clinically relevant image regions. Figs. 14, 15 and 16 shows heatmap results for COVID-19, normal, and viral pneumonia samples from the D_1, D_2, and D_{Mix} X-ray datasets using Grad-CAM visualizations. The model precisely detects infected lung regions in X-ray images, while no abnormalities are found in normal cases, highlighting its better interpretability and strong classification performance. Grad-CAM ensures that its visualization aligns with human expert analysis rather than making opaque or arbitrary predictions. Moreover, in clinical practice, AI-assisted diagnosis tools require explainability to be trusted by medical professionals. Compared to other explainability methods (e.g., LIME, SHAP), Grad-CAM provides pixel-wise activation maps directly overlaid on the original X-ray images [54]. This enables radiologists to verify whether the AI model focuses on clinically relevant areas, improving confidence in its predictions. This enhanced interpretability adds transparency and trust, essential for deploying AI-driven diagnostic models in real-world healthcare settings.

5. Discussions

This paper presents a deep learning model utilizing the EffiCOVID-Net architecture for detecting COVID-19 from chest X-ray images. The proposed model exhibited exceptional performance on D_1, D_2, and D_{Mix} datasets. Specifically, the model achieved an accuracy of 98.68% for 3-class classification and 99.06% for binary classification on the D_1 dataset. It also recorded an accuracy of 98.55% for 3-class classification and 99.78% for 2-class classification on the D_2 dataset. For the D_{Mix} dataset, the model achieved accuracies of 98.87% and 99.07% for 3-class and 2-class classifications, respectively. These results underscore the model's robustness and effectiveness in accurately classifying COVID-19 cases, thereby contributing valuable insights to medical imaging and disease detection.

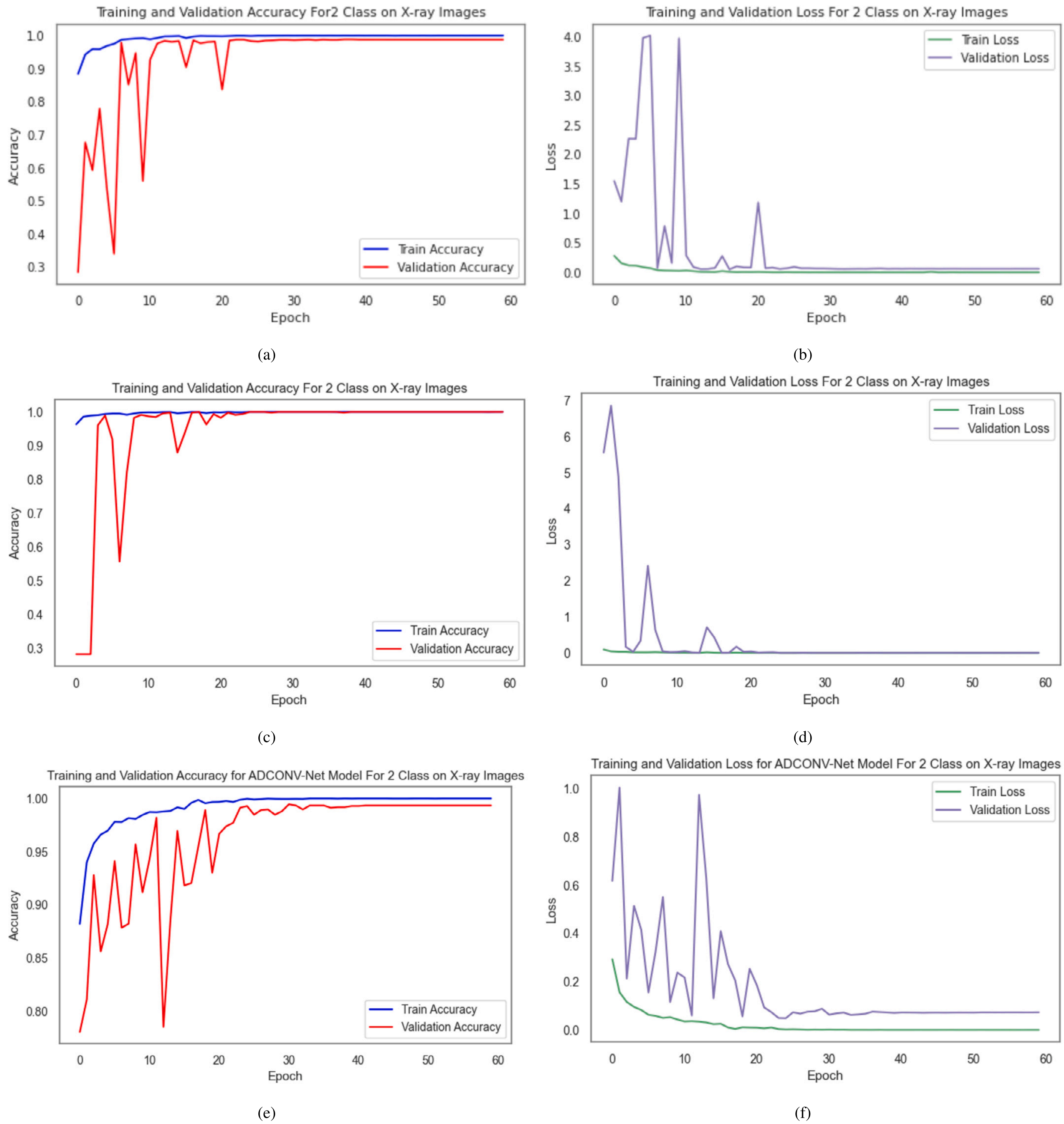
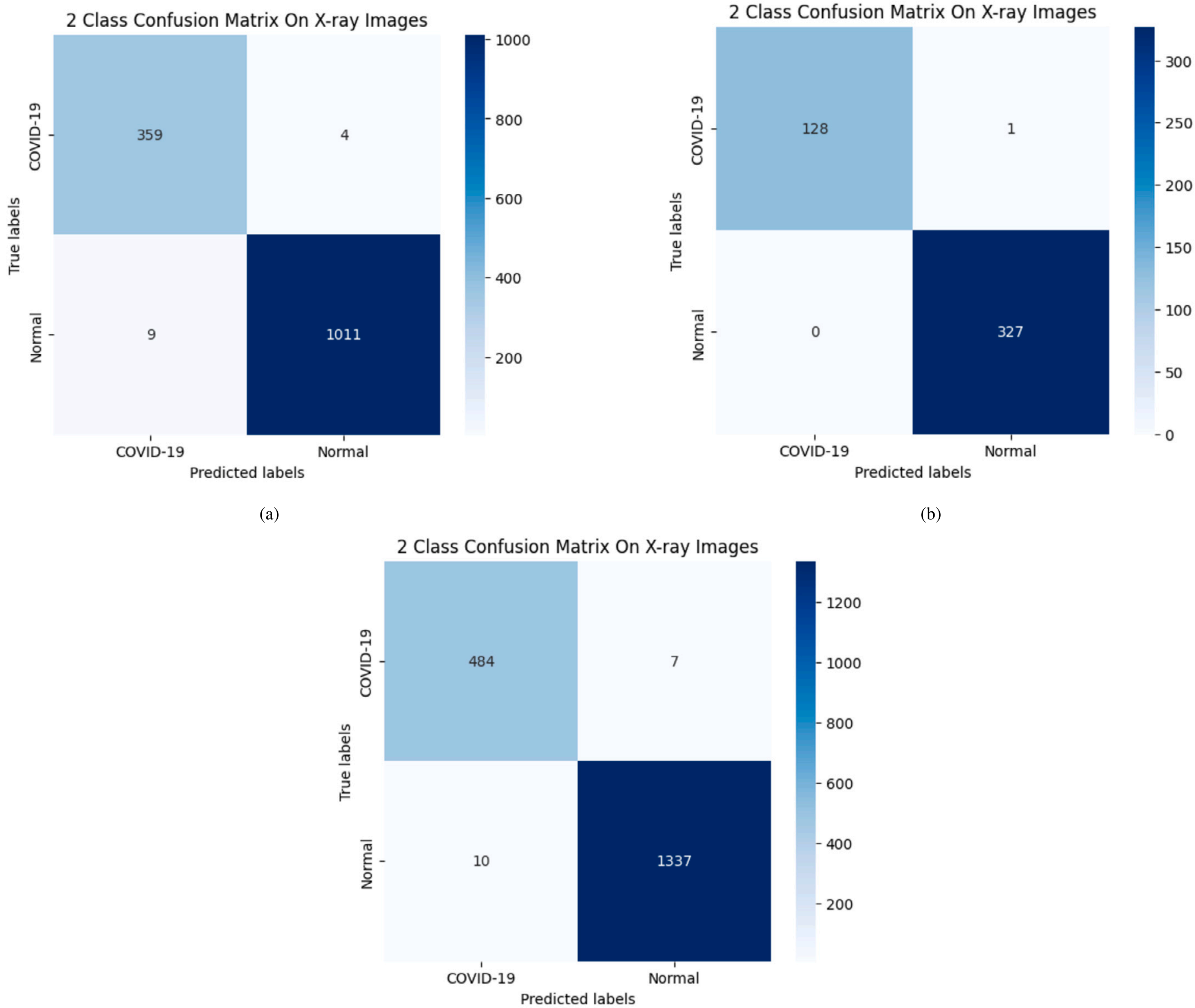


Fig. 11. Training Curves for 2 class classification (accuracy and loss vs. epoch) on (a) and (b) for D_1, (c) and (d) for D_2, and (e) and (f) for D_{Mix} dataset.

To validate the effectiveness of the proposed approach, we compared the performance of our EffiCOVID-net model with prior studies that used the same dataset for a similar classification task on the D_1 dataset (COVID-19 vs. viral pneumonia vs. normal), as displayed in Table 11. Additionally, Table 12 compares the results of three-class classification using chest X-ray radiograph images on the D_2 dataset (COVID-19 vs. viral pneumonia vs. normal) with the proposed approach versus existing methods, further validating the robustness of Efficovid-net across datasets. This comparison highlights how the proposed approach fares against other techniques in accuracy, robustness, and overall efficiency. It proves its superiority in classifying different respiratory conditions using the same dataset. To enhance the evaluation, given the common practice of merging datasets to increase sample sizes in chest X-ray analysis, we combined the D_1 and D_2 datasets into a unified D_{Mix} dataset. This ensures a more consistent and accurate assessment by comparing our model's performance against existing methods using a standardized dataset. Addressing variability introduced through dataset merging allows for a more comprehensive comparison, ensuring a consistent evaluation framework across different studies while maintaining fairness and robustness. As shown in Table 11, the proposed method achieved



(a) (b) (c)

Fig. 12. Confusion Matrices for 2 Class Classification on (a) D_1, (b) D_2, and (c) D_{Mix} dataset.

impressive results, attaining an accuracy of 98.68%, an F1-score of 98.29%, and a recall of 98.40% compared to alternative approaches. However, these metrics show that the proposed method is reliable on the D_1 dataset. However, the custom CNN model [55] did better than the proposed model, yielding a slightly higher precision score. Similarly, Table 12 indicates that the proposed method also delivered notable results on the D_1 dataset, achieving an accuracy of 98.55%, an F1-score of 98.49%, a recall of 98.23%, and precision of 98.78% when compared to similar existing approaches for chest radiogram image classification. These metrics emphasize the effectiveness of the proposed method in distinguishing between COVID-19, viral pneumonia, and normal cases, showcasing its robustness in clinical applications and reinforcing its potential utility in real-world diagnostic settings.

From Table 13, the comparison of two-class classification results across the D_1 and D_2 datasets, alongside existing methods, indicates that the proposed model achieved high accuracy compared to alternative approaches. This finding suggests that while the model exhibits overall strong performance, there are specific areas—particularly concerning accuracy—where improvements could be advantageous for both the D_1 and D_2 datasets. This highlights the necessity for ongoing enhancements and optimizations in the model's application to ensure robust performance across various datasets.

The performance difference between binary and multi-class classification is due to the complexity of distinguishing between multiple respiratory conditions. EffiCOVID-Net achieves high accuracy in binary classification by separating COVID-19 from non-COVID-19 cases due to distinct radiological patterns. However, in multi-class classification, the model must differentiate between COVID-19, viral pneumonia, and normal cases, increasing classification challenges because of overlapping visual features in chest X-ray images. This results in slightly lower performance in multi-class scenarios. To further understand these challenges, we analyzed misclassified cases and observed that certain pneumonia cases exhibited radiological similarities to COVID-19, making precise differentiation difficult.

Furthermore, the Grad-CAM technique was employed to visualize class activation heat maps for the proposed model. This method leverages gradients from target classes within a classification network to create a coarse localization map that highlights the significant regions

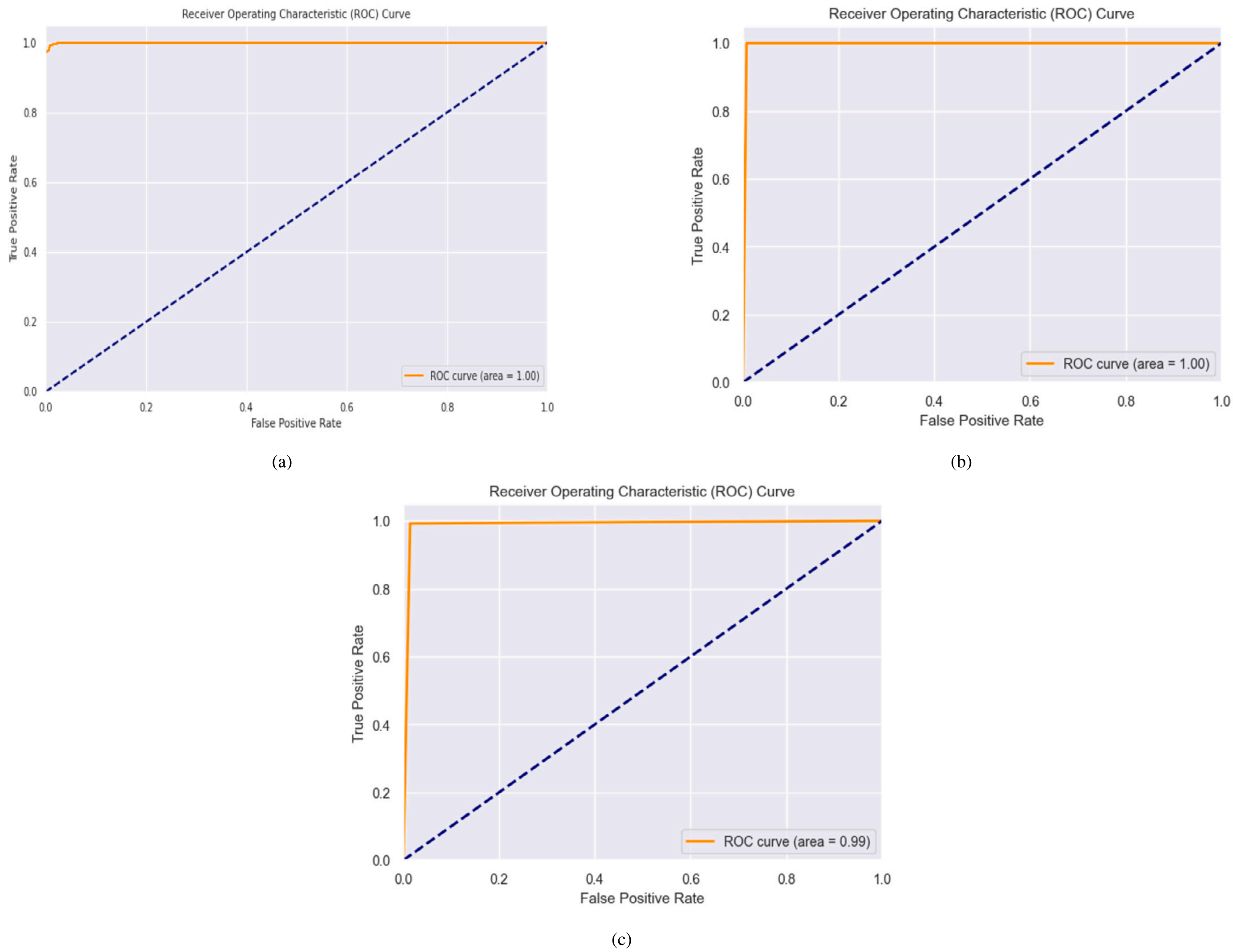
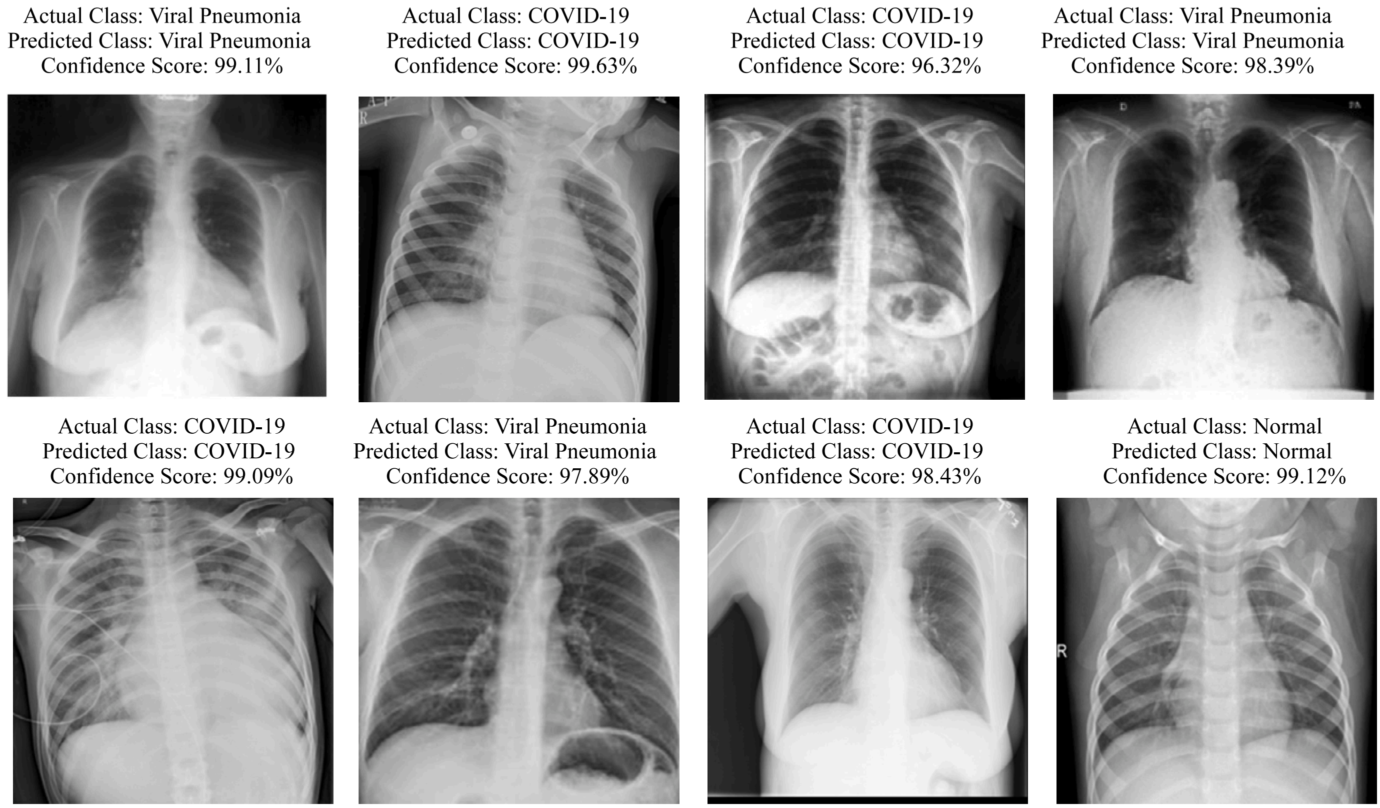


Fig. 13. AUC-ROC Curves for 2 class classification on (a) D_1, (b) D_2, and (c) D_{Mix} dataset.

of the image pertinent to specific class predictions. Figs. 14, 15, and 16 showcase the heatmap results for COVID-19, normal, and viral pneumonia samples drawn from the D_1, D_2, and D_{Mix} X-ray datasets, respectively. These visualizations provide insights into the critical areas the model focuses on during classification, revealing the features most indicative of each condition. Examining these heatmaps, we understand how the model differentiates between the various respiratory conditions based on lung images, significantly enhancing its predictions' interpretability. This ability to visualize class activations validates the model's decision-making process but also aids in identifying potential areas for improvement in future iterations. The chest X-rays alone have inherent limitations in differentiating COVID-19 from other respiratory conditions. Many lung infections, including bacterial and other viral pneumonia, exhibit overlapping radiographic patterns, making precise classification challenging. Although EffiCOVID-Net incorporates multi-scale feature learning to enhance discrimination, X-ray imaging alone cannot fully replace confirmatory diagnostic techniques like RT-PCR. This underscores the need to integrate additional clinical information or combine imaging modalities such as CT scans for more definitive diagnosis in complex cases.

The integration of EffiCOVID-Net into clinical workflows can alleviate the diagnostic burden on radiologists, particularly in resource-limited settings where expert radiologists may not always be available. The proposed model enables rapid and automated detection of COVID-19 from chest X-rays with high accuracy, significantly reducing the time required for manual interpretation. Given the high sensitivity and specificity of the model, it can serve as an effective pre-screening tool, allowing radiologists to focus on more complex or ambiguous cases. Prior studies have demonstrated that AI-assisted radiological assessments improve diagnostic efficiency while maintaining high accuracy [56,57]. Additionally, the lightweight nature of EffiCOVID-Net ensures low computational overhead, making it feasible for deployment in hospitals and clinics with limited computing resources. However, while the model enhances workflow efficiency, it is best utilized as a supportive tool rather than a standalone diagnostic method. Future real-world validation in clinical settings will further establish its practical impact on healthcare workflows. These advantages collectively reduce strain on healthcare systems by optimizing workflow efficiency, enabling faster diagnosis, and facilitating timely treatment interventions.

Overall, the proposed model features a streamlined architecture with fewer layers than other detection models, reducing computational complexity and training time. It accurately recognizes and categorizes both data types while mitigating overfitting in the training and testing phases. The use of balanced datasets can further improve classification performance. However, certain limitations persist, primarily due to the relatively small COVID-19 dataset and the lack of diversity in the X-ray images. Additionally, the model has not been extensively evaluated on patients across varying stages of infection. While prior studies have



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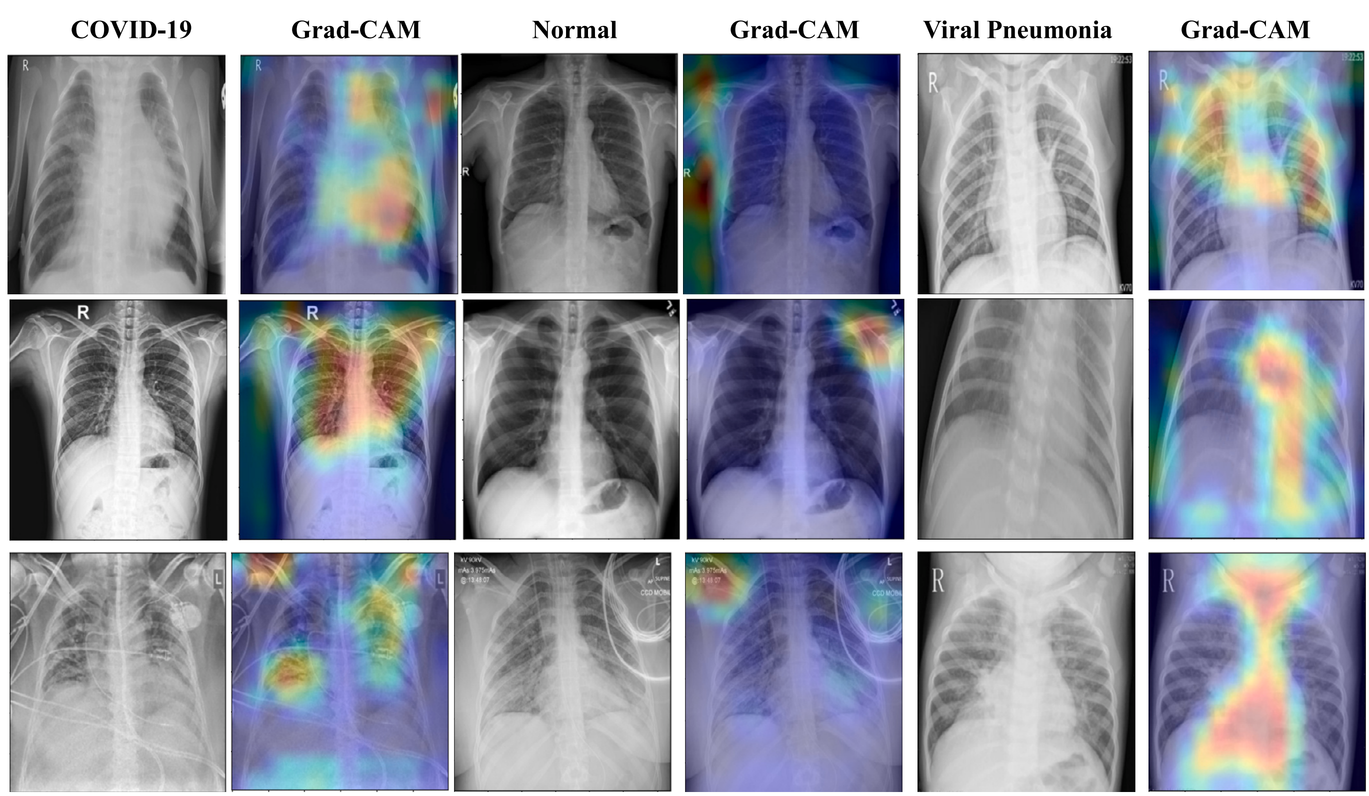
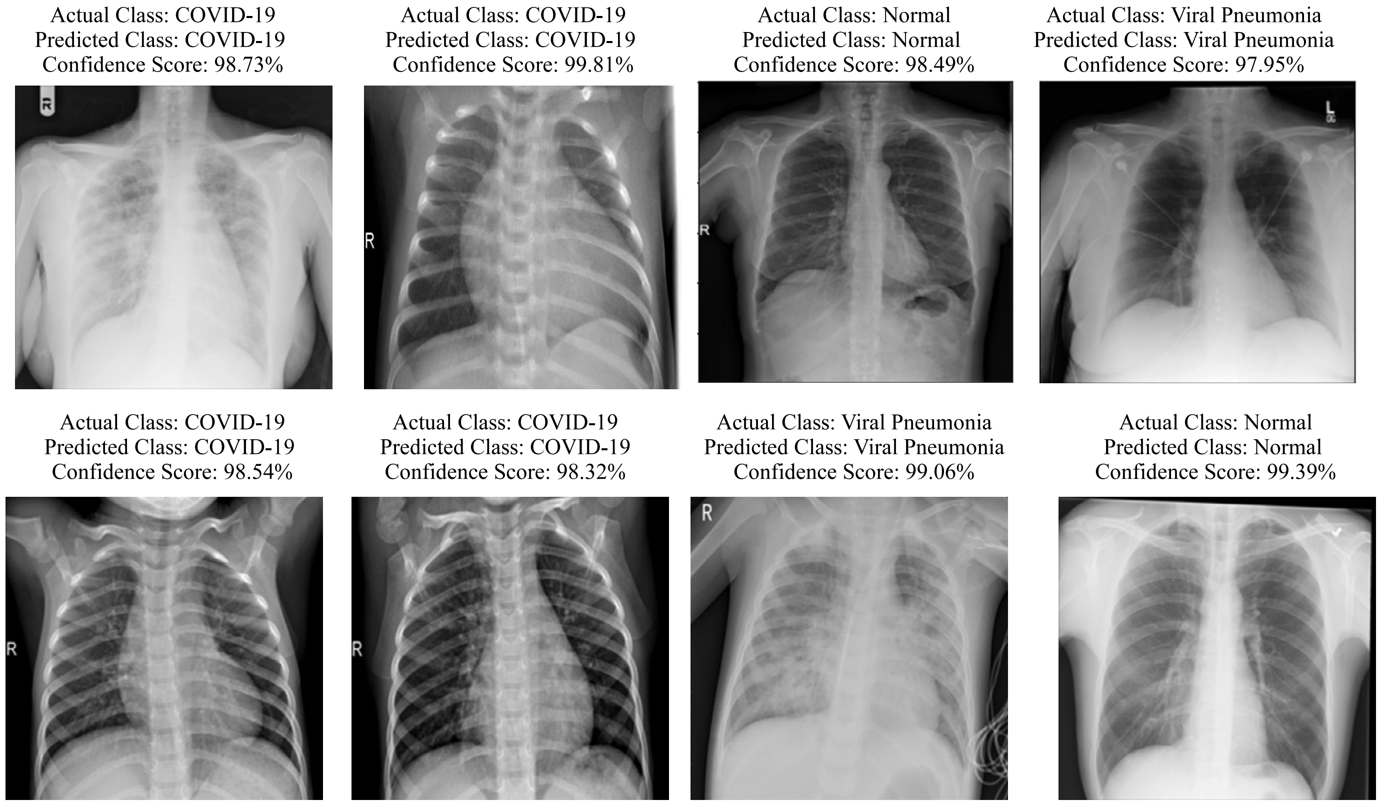
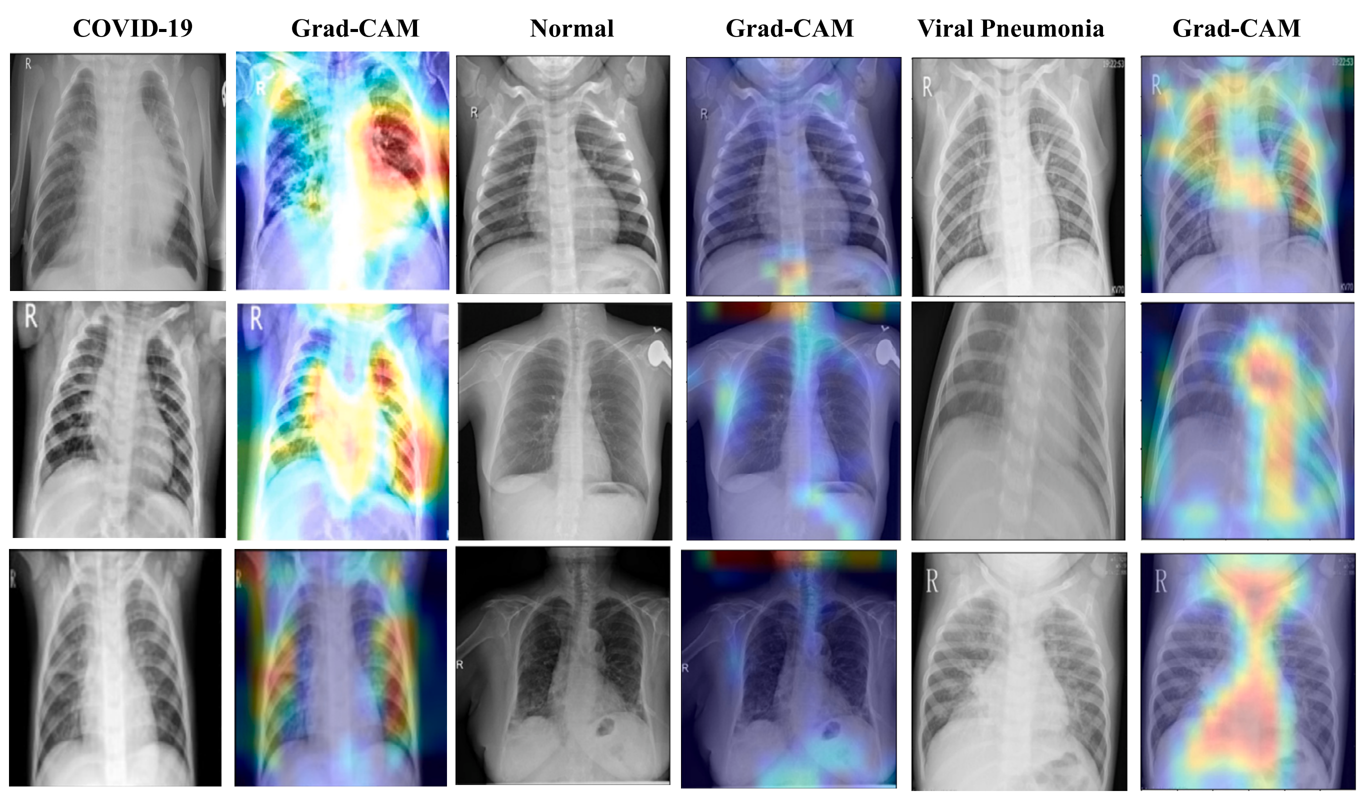


(b)

Fig. 14. Random Predictions of the EffiCOVID-Net Model Using (a) X-ray Images, and (b) GradCAM on the D_1 Dataset.



(a)



(b)

Fig. 15. Random Predictions of the EffiCOVID-Net Model (a) Xray Images, and (b) GradCAM on the D_2 Dataset.

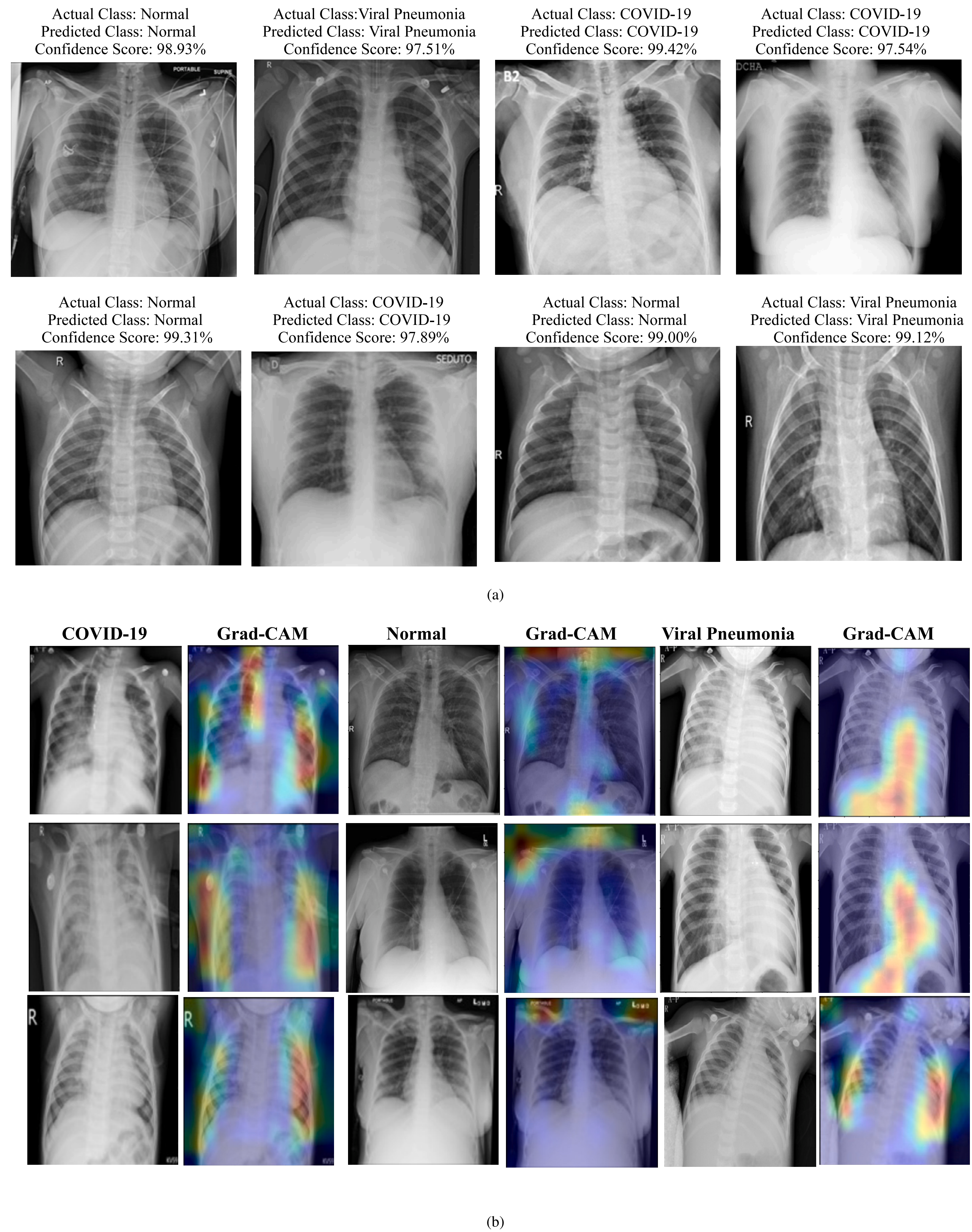


Fig. 16. Random Predictions of the EffiCOVID-Net Model Using (a) X-ray Images, and (b) GradCAM on the D_{Mix} Dataset.

Table 11
Performance comparison of 3 class chest radiography datasets (D_1) for COVID-19 detection.

Author	Method	Dataset	Recall (%)	Precision (%)	F1 Score (%)	Accuracy (%)
Chowdhury et al. [18]	GHE+TBH	COVID-19 radiography database	97.94	97.95	97.94	97.94
Ullah et al. [21]	CovidDet-Net	COVID-19 radiography database	96.66	97.00	96.82	98.40
Aslan et al. [24]	CNN	COVID-19 radiography database	98.26	98.16	98.20	98.14
Sakib et al. [25]	ResNet-50	COVID-19 radiography database	97.67	98.00	97.67	96.00
Oksuz et al. [26]	Ensemble-CVDNet	COVID-19 radiography database	97.78	97.43	97.61	98.30
Okolo et al. [55]	Custom CNN	COVID-19 radiography database	97.55	98.52	98.22	98.04
Proposed	EffiCOVID-Net	COVID-19 radiography database (D_1)	98.40	98.17	98.29	98.68

Table 12
Comparison of 3 class chest radiographs with existing approaches for COVID-19 detection.

Author	Method	Dataset	Recall (%)	Precision (%)	F1 Score (%)	Accuracy (%)
Ucar et al. [23]	Deep Bayes-SqueezeNet	COVID-19 radiography database	-	-	98.25	98.26
Wang et al. [28]	COVID-Net	COVIDx	93.33	93.56	-	93.30
Progga et al. [58]	VGG-16+HE	COVID-19 radiography database	98.00	98.00	98.00	-
Pham et al. [59]	Fine-tuned CNN	COVID-19 radiography database	95.45	98.55	96.90	97.59
Wang et al. [60]	CFW-Net	Public database	94.33	96.52	95.41	94.35
Bayram et al. [61]	HOG+CNN	COVID-19 radiography database	97.76	97.78	97.76	97.76
Proposed	EffiCOVID-Net	Curated Chest X-ray Image (D_2)	98.23	98.78	98.49	98.55
Proposed	EffiCOVID-Net	(D_{Mix})	98.67	98.71	98.68	98.87

Table 13
COVID-19 detection via 2-Class chest radiography: a Comparison with existing methods.

Author	Method	Number of Cases	Recall (%)	Precision (%)	F1 Score (%)	Accuracy (%)
Ozturk et al. [20]	DarkCovidNet	125 COVID-19 and 500 Normal	95.13	98.03	96.51	98.08
Narin et al. [62]	Deep CNN+ ResNet-50	50 COVID-19 and 50 Normal	91.8	76.5	83.5	96.1
Aayush et al. [63]	SARS-Net	358 COVID-19 and 8851 Normal	-	-	-	97.60
Khan et al. [64]	CoroNet	284 COVID-19 and 310 Normal	99.30	99.30	98.50	99.00
Ioannis et al. [65]	VGG-19	224 COVID-19 and 504 Normal	-	-	-	98.75
Proposed	EffiCOVID-Net	3,616 COVID-19 and 10,192 Normal (D_1)	99.00	98.57	98.79	99.06
Proposed	EffiCOVID-Net	1,281 COVID-19 and 3,217 Normal (D_2)	99.61	99.84	99.72	99.78
Proposed	EffiCOVID-Net	4,897 COVID-19 and 13,462 Normal (D_{Mix})	98.91	98.72	98.82	99.07

documented radiological patterns at different stages of disease progression, publicly available datasets with detailed severity annotations remain limited. Future work should focus on incorporating more diverse datasets and exploring multimodal approaches to enhance robustness and generalizability in real-world clinical settings.

6. Conclusions

The global COVID-19 pandemic has emphasized the urgent need for fast, affordable, and dependable diagnostic tools. This paper presents the EffiCOVID-Net model, which leverages multi-scale feature learning to detect COVID-19 from chest X-ray images accurately. The model has been evaluated using datasets D_1, D_2, and D_{Mix}, achieving impressive accuracies of 98.68% for D_1, 98.55% for D_2, and 98.87% for D_{Mix} in three-class classification, and 99.06% for D_1, 99.78% for D_2, and 99.07% for D_{Mix} in two-class classification. The EffiCOVID-Net model surpasses existing methods, offering a robust solution for improving COVID-19 diagnosis. Its implementation can ease the burden on healthcare systems by reducing diagnostic workload and helping limit the spread of the virus. Due to its end-to-end learning architecture, the proposed EffiCOVID-Net model can automatically detect COVID-19 from chest X-ray images, eliminating manual feature extraction. By adopting this approach, the workload for radiologists is reduced, and the risk of misdiagnosis is minimized.

Despite its strong performance, certain limitations must be acknowledged. The model's generalizability depends on dataset diversity, and its reliance on deep learning poses challenges related to interpretability. Additionally, while multiple datasets were used for validation, further evaluation of large-scale, real-world clinical data is necessary to confirm its robustness across different imaging conditions and demographic variations. To address these challenges, future research will explore integrating hybrid architectures that combine CNNs with attention mechanisms or transformers for improved robustness. Furthermore, explainable AI techniques will be incorporated to enhance the interpretability of predictions. Extending this work with incorporating CT scans and clinical data, could further refine diagnostic accuracy. Additionally, prospective validation in hospital settings will be crucial to assess real-world applicability and effectiveness. By addressing these research directions, the EffiCOVID-Net model has the potential to become an integral tool in radiological workflows, contributing to timely and accurate COVID-19 diagnosis and ultimately aiding in pandemic response efforts.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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